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MANAGEMENT, PUNE**

**AN EMPOWERED AUTONOMOUS INSTITUTE, AFFILIATED
TO SAVITRIBAI PHULE PUNE UNIVERSITY, PUNE**

Submitted for

TAE-I

in the subject

Database Management System (DBMS)

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**G H RAISONI COLLEGE OF ENGINEERING AND
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2. Database System Title

Bank Performance and Economic Indicator Database Management System

2.1 Introduction

The banking industry plays a critical role in economic development and generates substantial financial and operational information every year. The proposed database integrates bank-level performance measures with year-level economic indicators so that historical performance can be stored, queried, compared, and analysed systematically.

The design addresses the limitations of spreadsheet or flat-file storage, where bank information and year-level economic values may be repeated across many records. Such repetition can create update, insertion, and deletion anomalies and makes integrity enforcement difficult. The proposed relational design separates stable master data, bank-year performance data, indicator metadata, and economic observations into normalized relations.

2.2 Aim

To design a normalized relational database capable of storing, managing, and analysing banking performance together with macroeconomic indicators in a structured and scalable manner.

2.3 Objectives

- Design a conceptual database model using an Enhanced Entity–Relationship (EER) diagram.
- Identify enterprise entities, attributes, relationships, keys, cardinalities, and business constraints.
- Convert the conceptual design into a relational schema suitable for MySQL 8.0.
- Reduce redundancy and anomalies through normalization up to BCNF where the stated functional dependencies support it.
- Enforce entity integrity, referential integrity, and domain integrity.
- Integrate annual banking performance records with year-level economic indicators.
- Support comparative and analytical SQL queries across banks, sectors, indicators, and reporting years.

2.4 Scope

- Management of banking sector classifications.
- Storage of bank master records.
- Recording annual bank-level financial performance measures.
- Management of economic indicator metadata and yearly values.
- Year-wise, bank-wise, and sector-wise analysis.
- Combined analysis of bank performance and the economic environment.

2.5 Benefits

- Reduced data redundancy.
- Improved consistency and integrity.
- Reduced update, insertion, and deletion anomalies.
- Efficient SQL querying and reporting.
- Clear separation of master and measurement data.
- Scalability for additional banks, years, and indicators.

2.6 Expected Outcomes

- A complete conceptual EER model.
- A relational schema with keys and integrity constraints.
- A documented normalization path from raw structure to 3NF/BCNF.

2.7 Dataset Overview and Actual Data Profile

The current design is grounded in the uploaded banking workbook and the uploaded World Bank Development Indicators CSV. The banking workbook contains two sheets; the selected Sheet2 is the compact source used for this proposal.

Data Item	Observed Value
Banking workbook	Indian Banking Data (1).xlsx
Selected sheet	Sheet2
Selected banking rows	1,378
Distinct banks	116
Distinct sectors	3
Banking year range	2005–2020
Duplicate Bank–Year rows in selected sheet	0
Selected banking attributes	22
World Bank file	API_IND_DS2_en_csv_v2_1896 (1).csv
World Bank file structure	India indicator series in wide format, 1960–2025 columns

2.8 Data Provenance Note

The uploaded World Bank CSV is a World Development Indicators file and can support indicators such as GDP growth and inflation. The banking workbook also contains variables such as repo, government expenditure, and other derived measures. These variables should not automatically be described as World Bank indicators unless their original provenance is separately verified. Accordingly, this report treats the uploaded WDI file and the banking workbook as distinct source inputs and avoids claiming that every economic variable in the banking workbook originated from the World Bank.

2.9 Assumptions

1. Each bank belongs to one sector classification.
2. A bank can have multiple yearly financial performance records.
3. A bank has at most one performance record for a given reporting year.
4. Each economic indicator type can have a value for a given reporting year.
5. Primary keys are unique and non-null.
6. Foreign keys reference valid parent records.
7. Year-level economic indicators are analysed against bank performance using the common reporting year.

3. Problem Statement

3.1 Existing Problems

- Data redundancy: bank and economic information can be repeated across multiple annual records.
- Update anomaly: changing a bank's sector or descriptive information in repeated rows can create inconsistency.
- Insertion anomaly: adding a new sector or indicator is awkward when data is stored only inside combined observations.
- Deletion anomaly: deleting the last performance observation can unintentionally remove descriptive information.
- Weak integrity enforcement: flat files do not naturally enforce foreign-key relationships.
- Limited scalability: increasingly large files become harder to maintain and query reliably.
- Inefficient reporting: multi-year comparisons require repeated filtering and manual processing.

3.2 Proposed Solution

- A normalized relational database is proposed in which sector information, bank master information, bank-year financial performance, indicator metadata, and yearly economic observations are stored separately and linked through primary and foreign keys.

3.3 Enterprise Entities

Entity	Purpose
SECTOR	Stores banking sector classification.
BANK	Stores master information about banks.
FINANCIAL PERFORMANCE	Stores annual bank-level financial performance metrics.
INDICATOR_TYPE	Stores economic indicator metadata.
ECONOMIC_INDICATOR	Stores yearly values for economic indicators.

3.4 Business Rules

1. Every bank belongs to exactly one sector.
2. One sector can contain many banks.
3. One bank can have many yearly financial performance records.
4. A bank cannot have more than one performance record for the same year.
5. Every economic indicator value belongs to one indicator type.
6. An indicator type can have many yearly observations.
7. Financial performance and economic indicators are analytically compared by reporting year; no direct foreign key is required between them.
8. Primary keys must be unique and non-null.
9. Foreign keys must reference valid parent rows.
10. Numeric financial and economic measures must use suitable numeric domains.

3.5 Primary Operational Transactions

Transaction	Description
Add Sector	Insert a new banking sector classification.
Update Sector	Modify sector information.
Add Bank	Register a new bank.
Update Bank	Modify bank master information.
Delete Bank	Remove obsolete bank records subject to referential rules.
Add Financial Performance	Store a bank's annual performance record.
Update Financial Performance	Correct or modify a bank-year record.
Delete Financial Performance	Remove an incorrect performance record.
Add Indicator Type	Register a new economic indicator.
Add Economic Indicator	Store an indicator value for a year.
Update Economic Indicator	Modify an indicator observation.
Generate Reports	Produce bank-wise, year-wise, sector-wise, and combined analytical reports.

3.6 Users

- Banking and financial analysts.
- Financial researchers and economists.
- Regulatory or policy-analysis users.
- Database administrators.
- Academic researchers.

4. System Requirements

4.1 Functional Requirements

- Create, update, and retrieve sector records.
- Register and manage bank master records.
- Insert, update, delete, and retrieve annual financial performance.
- Create and manage indicator metadata.
- Store and retrieve year-wise economic indicator values.
- Generate bank-wise, sector-wise, year-wise, and combined reports.
- Support joins, aggregation, filtering, sorting, and trend analysis.

4.2 Non-Functional Requirements

Requirement	Description
Reliability	Maintain accurate and internally consistent records.
Scalability	Accommodate new banks, years, and indicators.
Maintainability	Use normalized relations and clear constraints.
Security	Support controlled database access and privileges.
Performance	Use primary keys, foreign keys, and suitable indexes.
Availability	Provide dependable access to stored information.

4.3 Hardware Requirements

Component	Minimum / Suggested Specification
Processor	Intel Core i3 or higher
RAM	8 GB
Storage	20 GB available space
Display	1366 × 768 or higher
Network	Internet connection for documentation/data-source access

4.4 Software Requirements

Software	Selection
Operating System	Windows 10 / Windows 11
Database Engine	MySQL 8.0
Database IDE	MySQL Workbench
Documentation	Microsoft Word
Diagram Tool	draw.io / equivalent diagramming software

Software	Selection
Implementation Language	SQL

4.5 Expected Outputs

- Bank performance reports.
- Sector performance reports.
- Year-wise financial reports.
- Economic indicator reports.
- Comparative banking analysis.
- Financial trend analysis.

5. Literature & System Reference

5.1 Introduction

The literature and system reference review connects the proposed database to established banking performance analysis, relational database design, and macroeconomic data practices.

5.2 Banking Performance Analysis

Bank performance is commonly studied using profitability, efficiency, liquidity, capital, asset-quality, and concentration/diversification measures. The proposed schema therefore captures selected measures such as Net Interest Income, Return on Assets, Deposit Ratio, Loan Ratio, Equity Ratio, Asset Growth, staffing, risk measures, HHI, and diversification.

5.3 Economic Indicators

Macroeconomic conditions can be compared with bank performance on a yearly basis. The uploaded World Bank Development Indicators file contains India indicator series in a wide format. The relational design converts selected indicator series into an indicator metadata relation and a year-value relation so that additional indicator types can be added without adding new columns to the core economic table.

5.4 Relational Database Model

The relational model stores information in relations connected through keys. The proposed design uses separate master and measurement relations to reduce duplication, enforce integrity, and support SQL joins.

5.5 Normalization

Normalization is used to reduce redundancy and prevent update, insertion, and deletion anomalies. The design demonstrates a path from the raw combined structure through 1NF, 2NF, 3NF and, where the stated determinants are candidate keys, BCNF.

5.6 Research / Technical References

Reference	Relevance to Project
Berger & Humphrey (1997)	Provides a broad review of efficiency analysis in financial institutions.
Athanasoglou, Brissimis & Delis (2008)	Examines bank-specific, industry-specific, and macroeconomic determinants of bank profitability.
Demirgüç-Kunt & Huizinga (1999)	Studies determinants of commercial bank interest margins and profitability.
Elmasri & Navathe, Fundamentals of Database Systems	Provides the relational design, keys, dependencies, and normalization foundation.
Oracle, MySQL 8.0 Reference Manual	Provides the target RDBMS implementation reference for constraints and SQL operations.
World Bank, World Development Indicators	Provides an official source for macroeconomic indicator series used as an external data source.

6. Conceptual Schema Design

6.1 Introduction

The conceptual schema identifies the enterprise entities, attributes, keys, relationships, cardinalities, and business constraints independently of physical SQL implementation.

6.2 Enterprise Entities and Attributes

Entity	Attributes
SECTOR	SectorID (PK); SectorName
BANK	BankID (PK); BankName; SectorID (FK)
FINANCIAL_PERFORMANCE	PerformanceID (PK); BankID (FK); Year; NII; ROA; DepositRatio; ProvisionNPA; TotalStaff; ROARisk; NIIOperatingIncome; LoanRatio; EquityRatio; LogTotalAssets; AssetGrowth; ROARisk5Year; HHI_NII; DiversificationIndex
INDICATOR_TYPE	IndicatorTypeID (PK); IndicatorName; IndicatorCode
ECONOMIC_INDICATOR	IndicatorTypeID (PK/FK); Year (PK); IndicatorValue

6.3 Candidate / Alternate Keys

Entity	Candidate / Alternate Key	Reason
SECTOR	SectorName	Sector name is intended to be unique.
BANK	BankName	Bank name is intended to be unique in the master data.
FINANCIAL_PERFORMANCE	(BankID, Year)	One performance record per bank per reporting year.

Entity	Candidate / Alternate Key	Reason
INDICATOR_TYPE	IndicatorName; IndicatorCode	Each is intended to uniquely identify an indicator type.
ECONOMIC_INDICATOR	(IndicatorTypeID, Year)	One value per indicator type per year.

6.4 Primary Keys

Entity	Primary Key
SECTOR	SectorID
BANK	BankID
FINANCIAL_PERFORMANCE	PerformanceID
INDICATOR_TYPE	IndicatorTypeID
ECONOMIC_INDICATOR	(IndicatorTypeID, Year)

6.5 Foreign Keys

Child Entity	Foreign Key	Parent Entity
BANK	SectorID	SECTOR(SectorID)
FINANCIAL_PERFORMANCE	BankID	BANK(BankID)
ECONOMIC_INDICATOR	IndicatorTypeID	INDICATOR_TYPE(IndicatorTypeID)

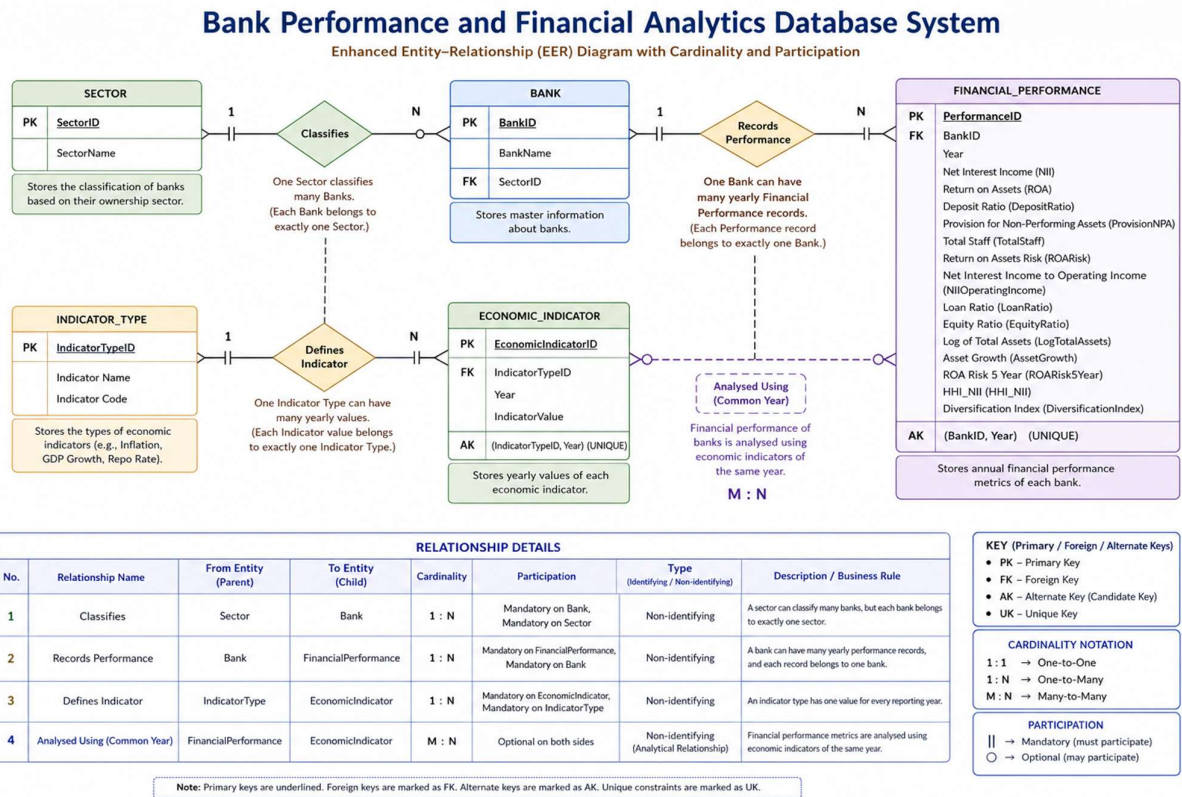
6.6 Relationship Descriptions

Relationship	Cardinality	Participation / Rule
SECTOR — Classifies — BANK	1 : N	Every bank belongs to exactly one sector; a sector can classify many banks.
BANK — Records Performance — FINANCIAL_PERFORMANCE	1 : N	Each performance record belongs to one bank; a bank can have many yearly records.
INDICATOR_TYPE — Defines Indicator — ECONOMIC_INDICATOR	1 : N	Each economic value belongs to one indicator type; an indicator type can have many yearly values.
FINANCIAL_PERFORMANCE ↔ ECONOMIC_INDICATOR	M : N (analytical)	Compared using common Year; no direct FK is enforced. This is an analytical/derived relationship.

6.7 Business Constraints

1. Every bank belongs to exactly one sector.
2. One sector may contain multiple banks.
3. One bank cannot have two performance records for the same year.
4. One indicator type can have at most one value for a given year.
5. Primary keys are unique and non-null.
6. Foreign keys reference valid parent rows.
7. Year is stored as a numeric reporting-year domain.
8. Financial and economic measurements use appropriate numeric types.

6.8 EER Diagram



7. Relational Schema Design

7.1 Conversion of EER to Relations

The conceptual model is converted into five principal relations. Master entities are separated from bank-year and indicator-year measurements so that descriptive data is not repeated unnecessarily.

7.2 SECTOR

Attribute	Data Type	Constraint	Description
SectorID	INT	PK, NOT NULL	Unique sector identifier
SectorName	VARCHAR(30)	NOT NULL, UNIQUE	Sector classification

7.3 BANK

Attribute	Data Type	Constraint	Description
BankID	INT	PK, NOT NULL	Unique bank identifier
BankName	VARCHAR(150)	NOT NULL, UNIQUE	Bank name
SectorID	INT	FK, NOT NULL	References SECTOR

7.4 FINANCIAL_PERFORMANCE

Attribute	Data Type	Constraint	Description
PerformanceID	INT	PK	Surrogate identifier
BankID	INT	FK, NOT NULL	Bank reference
Year	SMALLINT	NOT NULL	Reporting year
NetInterestIncome	DECIMAL(12,4)		NII
ReturnOnAssets	DECIMAL(10,4)		ROA
DepositRatio	DECIMAL(10,4)		Deposit ratio
ProvisionNPA	DECIMAL(10,4)		Provision for NPA
TotalStaff	INT		Total staff
ROARisk	DECIMAL(10,6)		ROA risk
NIIOperatingIncome	DECIMAL(12,6)		NII to operating income
LoanRatio	DECIMAL(10,4)		Loan ratio
EquityRatio	DECIMAL(10,4)		Equity ratio
LogTotalAssets	DECIMAL(12,6)		Log total assets
AssetGrowth	DECIMAL(10,4)		Asset growth
ROARisk5Year	DECIMAL(10,6)		Five-year ROA risk
HHI_NII	DECIMAL(12,6)		HHI based on NII
DiversificationIndex	DECIMAL(14,6)		Diversification measure

7.5 INDICATOR_TYPE

Attribute	Data Type	Constraint	Description
IndicatorTypeID	INT	PK	Unique indicator type identifier
IndicatorName	VARCHAR(100)	NOT NULL, UNIQUE	Human-readable indicator name
IndicatorCode	VARCHAR(30)	NOT NULL, UNIQUE	Source/standard indicator code where available

7.6 ECONOMIC_INDICATOR

Attribute	Data Type	Constraint	Description
IndicatorTypeID	INT	PK, FK	Indicator reference
Year	SMALLINT	PK	Reporting year
IndicatorValue	DECIMAL(15,6)	NOT NULL	Value for indicator and year

8. Database Normalisation

8.1 Purpose

Normalization reduces redundancy and prevents update, insertion, and deletion anomalies by decomposing the raw structure into relations whose attributes depend on appropriate determinants.

8.2 Raw / UNF Structure

```
BANKING_RAW(  
  Sector,  
  Bank,  
  Year,  
  NII,  
  ROA,  
  DepositRatio,  
  ProvisionNPA,  
  TotalStaff,  
  Inflation,  
  ROARisk,  
  NIIOperatingIncome,  
  LoanRatio,  
  EquityRatio,  
  LogTotalAssets,  
  AssetGrowth,  
  GDPGrowth,  
  GFCE,  
  ROARisk5Year,  
  GovernmentExpenditure,  
  RepoRate,  
  HHI_NII,  
  DiversificationIndex  
)
```

8.3 Functional Dependencies

FD	Functional Dependency	Meaning
FD1	Bank → Sector	A bank belongs to one sector.
FD2	(Bank, Year) → bank-specific performance attributes	A bank-year determines the selected performance measurements.
FD3	Year → year-level economic indicators	For indicators that are genuinely year-level and constant across bank records, the year determines their value.

FD3 must be applied only to indicators whose values are verified as year-level and common across all bank observations. It should not be assumed for every economic-looking field without provenance and consistency checks.

8.4 UNF → 1NF

The first step is to ensure atomic attributes and one observation per bank-year record. The compact raw structure contains scalar fields rather than repeating groups.

```
BANKING_1NF(  
  Bank,
```

```

Sector,
Year,
NII,
ROA,
DepositRatio,
ProvisionNPA,
TotalStaff,
ROARisk,
NIIOperatingIncome,
LoanRatio,
EquityRatio,
LogTotalAssets,
AssetGrowth,
ROARisk5Year,
HHI_NII,
DiversificationIndex,
Inflation,
GDPGrowth,
GFCE,
GovernmentExpenditure,
RepoRate
)

```

Candidate key for the current bank-year observations: (Bank, Year).

8.5 1NF → 2NF

With (Bank, Year) as the composite candidate key, attributes that depend only on Bank or only on Year represent partial dependencies. Sector is a bank-level attribute. Verified year-level indicators are year-level attributes.

Bank → Sector

Year → Inflation, GDPGrowth, GFCE,
GovernmentExpenditure, RepoRate

Decomposition:

```

BANK_2NF(
  Bank,
  Sector
)

```

```

ECONOMIC_YEAR_2NF(
  Year,
  Inflation,
  GDPGrowth,
  GFCE,
  GovernmentExpenditure,
  RepoRate
)

```

```

FINANCIAL_PERFORMANCE_2NF(
  Bank,
  Year,
  NII,
  ROA,
  DepositRatio,
  ProvisionNPA,
  TotalStaff,
  ROARisk,
  NIIOperatingIncome,
  LoanRatio,
  EquityRatio,
  LogTotalAssets,
  AssetGrowth,
  ROARisk5Year,
)

```

```

        HHI_NII,
        DiversificationIndex
    )

```

8.6 2NF → 3NF

The 3NF design separates descriptive master data from measurement data. Sector describes a bank, while year-level economic values describe the reporting year/indicator rather than an individual bank.

```

SECTOR(
    SectorID,
    SectorName
)

BANK(
    BankID,
    BankName,
    SectorID
)

FINANCIAL_PERFORMANCE(
    PerformanceID,
    BankID,
    Year,
    performance attributes...
)

INDICATOR_TYPE(
    IndicatorTypeID,
    IndicatorName,
    IndicatorCode
)

ECONOMIC_INDICATOR(
    IndicatorTypeID,
    Year,
    IndicatorValue
)

```

8.7 3NF → BCNF

BCNF requires every determinant of a non-trivial functional dependency to be a superkey. Under the proposed uniqueness constraints, the following determinants are keys or candidate keys.

Relation	Determinant	Dependent Attributes	BCNF Basis
SECTOR	SectorID	SectorName	SectorID is the PK.
BANK	BankID	BankName, SectorID	BankID is the PK.
BANK	BankName	BankID, SectorID	BankName is declared UNIQUE, hence a candidate key.
FINANCIAL_PERFORMANCE	PerformanceID	BankID, Year, performance attributes	PerformanceID is the PK.
FINANCIAL_PERFORMANCE	(BankID, Year)	PerformanceID and performance attributes	Declared UNIQUE; candidate key under the business rule.
INDICATOR_TYPE	IndicatorTypeID	IndicatorName, IndicatorCode	IndicatorTypeID is the PK.
INDICATOR_TYPE	IndicatorName / IndicatorCode	Other indicator metadata	Declared UNIQUE; candidate keys if uniqueness is valid in the domain.
ECONOMIC_INDICATOR	(IndicatorTypeID, Year)	IndicatorValue	Composite PK is the determinant.

8.8 Integrity Constraints

Entity Integrity

- Every relation has a primary key.
- Primary key values are unique and cannot be NULL.

Referential Integrity

BANK.SectorID
→ SECTOR.SectorID

FINANCIAL_PERFORMANCE.BankID
→ BANK.BankID

ECONOMIC_INDICATOR.IndicatorTypeID
→ INDICATOR_TYPE.IndicatorTypeID

Domain Integrity

- Year is stored as a numeric reporting-year domain.
- TotalStaff is a non-negative integer.
- Bank and sector names are non-null.
- Indicator values are numeric.
- Financial ratios and measures use suitable DECIMAL domains.
- UNIQUE(BankID, Year) prevents duplicate bank-year performance records.

8.9 Normalisation Summary

Stage	Result
UNF	Combined bank, performance, and year-level economic attributes appear in one row structure.
1NF	Atomic attributes and one bank-year observation per row.
2NF	Partial dependencies on Bank and Year are separated.
3NF	Master/descriptive dependencies are separated into dedicated relations.
BCNF	Every stated determinant in the final relations is a candidate key/superkey under the declared uniqueness assumptions.

9. Target RDBMS Software

9.1 Selected RDBMS: MySQL 8.0

MySQL 8.0 is selected as the target RDBMS. It supports relational tables, primary and foreign keys, unique constraints, check constraints, indexes, SQL querying, transactions, and access privileges.

9.2 Why MySQL?

- Widely used relational database engine with strong SQL support.
- Suitable for academic implementation and demonstration.

- Supports primary keys, foreign keys, UNIQUE, NOT NULL, and CHECK constraints.
- Supports indexing and query optimization.
- Supports transaction control and access privileges.
- Compatible with MySQL Workbench for schema development and SQL execution.

9.3 Features Used in This Project

Feature	Purpose
CREATE DATABASE	Create the project database.
CREATE TABLE	Create the relational tables.
PRIMARY KEY	Enforce entity uniqueness.
FOREIGN KEY	Maintain referential integrity.
UNIQUE	Prevent duplicate candidate-key values.
NOT NULL	Prevent missing mandatory values.
CHECK	Enforce domain rules where appropriate.
INSERT / UPDATE / DELETE	Manipulate records.
SELECT	Retrieve records.
JOIN	Combine related relations.
GROUP BY	Produce aggregate reports.
ORDER BY	Sort analytical output.
INDEX	Improve retrieval performance.

9.4 SQL Operations

Category	Representative Operations
DDL	CREATE DATABASE, CREATE TABLE, ALTER TABLE, DROP TABLE
DML	INSERT, UPDATE, DELETE
DQL	SELECT, WHERE, GROUP BY, HAVING, ORDER BY
DCL	GRANT, REVOKE
TCL	COMMIT, ROLLBACK, SAVEPOINT

9.5 Expected SQL Queries

- Retrieve all banks in a selected sector.
- Display annual performance for a selected bank.
- Compare ROA across years and sectors.
- Retrieve economic indicators for a selected year.
- Join bank performance with economic indicators by year.
- Calculate average profitability by sector and year.
- Generate trend reports.
- Identify banks with high/low performance measures.

9.6 Future Enhancements

- Branch management.
- Customer and account modules.
- Loan and repayment modules.
- Employee management.
- Transaction processing.
- Business intelligence dashboards.
- Machine-learning-based performance prediction.

9.7 Conclusion

MySQL 8.0 provides an appropriate implementation platform for the proposed normalized schema and can support the SQL creation, data import, analytical querying, and performance-tuning activities expected.

10. References

Books

1. Elmasri, R., & Navathe, S. B. (2021). Fundamentals of Database Systems (7th ed.). Pearson.
2. Coronel, C., & Morris, S. (2019). Database Systems: Design, Implementation, & Management (13th ed.). Cengage Learning.

Official Documentation and Data Sources

1. Oracle Corporation. MySQL 8.0 Reference Manual.
2. World Bank. World Development Indicators (uploaded India indicator CSV).

Research Papers

1. Berger, A. N., & Humphrey, D. B. (1997). Efficiency of Financial Institutions: International Survey and Directions for Future Research. European Journal of Operational Research, 98(2), 175–212.

[Datasets 1:](#)

[Datasets 2](#)